

Statistical Techniques for Computer Applications

Unit No 1: Introduction to Statistics

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Syllabus

Unit 1- Introduction to Statistics

- Meaning and Definition of Statistics, Scope and Importance of Statistics in Computer Applications

- Types of data: Primary and Secondary data, Discrete and Continuous data, Qualitative and Quantitative Data

- Classification and Presentations of Data:

Types of Data Classification and Importance of Data Classification

- Graphical Presentation of data:

Presentation of Frequency Distribution by Histogram, Frequency Polygon, Frequency Curve

- Diagrammatic Representation using Multiple and Percentage Bar Diagrams



Meaning and Definition of Statistics



Meaning

Statistics is broadly defined as the science of collecting, organizing, analyzing, interpreting, and presenting data. It transforms raw numbers into meaningful information.

Definition

By Croxton and Cowden:

"Statistics may be defined as the collection, presentation, analysis, and interpretation of numerical data."

Case Study

An IT company launches an e-commerce website. After launch, users start complaining that the website is slow and sometimes crashes. The management asks the project team to analyze the problem using statistical methods.

Definition

Collection of Data

The project team collects numerical data from the web server for 30 days, such as:

Page load time (in seconds)

Number of users per hour

Number of server crashes per day

CPU and memory usage (%)

e.g. This step represents collection of numerical data.

Presentation of Data

The collected data is presented in:

Tables (daily page load time)

Line graphs (traffic vs time)

Bar charts (number of crashes per day)

e.g. A line graph shows that page load time increases sharply between 7 pm – 10 pm.

Definition

Analysis of Data

Using statistical techniques, the MCA students:

Calculate average (mean) page load time

Find peak traffic hours

Identify correlation between users and crashes

Use variance to check performance consistency

□ *Example:*

Average page load time = **4.1 seconds**

High positive correlation between **users and crashes**

Definition

Interpretation of Data

- ☐ Based on analysis, conclusions are drawn:
- ☐ Website performance degrades during peak hours
- ☐ Server capacity is insufficient
- ☐ More traffic leads to more crashes

Decision taken:

- ☐ Upgrade server hardware
- ☐ Use load balancing
- ☐ Optimize database queries

**This case study proves Croxton and Cowden's definition:
Statistics involves the collection, presentation, analysis,
and interpretation of numerical data to solve real-world
computer problems.**

Importance of Studying Statistical Techniques

1. Foundation for Data Science & AI

Statistical techniques help in:

- Understand data patterns
 - Do hypothesis testing
 - Measure accuracy, risk, and uncertainty
- Without statistics, AI models become *black boxes*.

2. Core Requirement for Machine Learning

- Algorithms like:
 - Linear & Logistic Regression
 - Naïve Bayes
 - Clustering & Classification
- are purely statistical at heart.
- Statistics explains why an algorithm works, not just *how to code it*.

Importance of Studying Statistical Techniques

3. Better Decision-Making in Software Projects

In real IT projects, students must:

- Analyze user behavior
- Evaluate system performance
- Compare algorithms

Statistical tools help in evidence-based decisions, not guesswork.

4. Handling Real-World Uncertainty

In the IT world:

- Data is incomplete
- Systems fail
- Results vary

Statistics trains students to manage uncertainty logically, a key professional skill.

Importance of Studying Statistical Techniques

5. Research & Higher Studies Advantage

For:

- MCA projects
- Research papers
- PhD / M.Tech / MS

Statistical techniques are essential for data analysis, validation, and conclusions.

6. Industry Demand & Employability

Companies expect MCA graduates to:

- Interpret reports
- Analyze trends
- Validate outputs

Statistics boosts job readiness and interview confidence.

Importance of Studying Statistical Techniques

7. Improves Logical & Analytical Thinking

- Statistics sharpens:
- Critical thinking
- Problem-solving
- Quantitative reasoning

These skills help not only in tech roles but also in **managerial and consulting positions**.

Scope and Importance of Statistics in Computer Applications



Scope

Where it is used

“In which fields?”

Areas of application

Importance

Why it is useful

“For what purpose?”

Benefits and value

Scope of Statistics in Computer Applications



Scope of Statistics in Computer Applications

1. Data Collection and Data Analysis

Statistics is used to collect, organize, and analyze large amounts of data.

Example:

- A website collects user details such as age, location, and time spent on the site. Statistics helps summarize this data and identify user behavior patterns.
- An institute collects student marks. Statistics helps find pass percentage and average marks.

Scope of Statistics in Computer Applications

2. Data Science and Analytics

Statistics helps in finding trends and patterns from data.

Example :

- An online store analyzes monthly sales to identify the best-selling product.
- A company analyzes customer feedback ratings to improve services.

Scope of Statistics in Computer Applications

3. Machine Learning and Artificial Intelligence

Statistical techniques are used to train and evaluate AI models.

Example :

- Classifying emails as spam or non-spam using probability.
- Predicting student performance based on attendance and marks.

Scope of Statistics in Computer Applications

4. Software Testing and Quality Assurance

Statistics helps in measuring software reliability and quality.

Example :

- Calculating defect rate after testing a program multiple times.
- Comparing bug counts of two software versions.

Scope of Statistics in Computer Applications

5. Database Management and Data Mining

Statistics helps extract useful information from large databases.

Example :

- Grouping customers based on transaction history.
- Identifying frequently accessed database records.

Scope of Statistics in Computer Applications

6. Network and System Performance Analysis

Statistics is used to evaluate system efficiency and performance.

Example :

- Analyzing average server response time.
- Measuring network downtime frequency.

Scope of Statistics in Computer Applications

7. Decision Support Systems

Statistics supports decision-making in computer applications.

Example :

- Selecting the best software based on user ratings.
- Choosing hardware configuration based on performance data.

Scope of Statistics in Computer Applications

8. Research and MCA Project Work

Statistics is essential for analyzing and validating project results.

Example :

- Comparing execution time of two algorithms.
- Analyzing survey data collected for an MCA project.

Scope of Statistics in Computer Applications

9. Cyber Security and Risk Analysis

Statistics helps in identifying abnormal patterns and risks.

Example :

- Detecting unusual login attempts.
- Analyzing failed login frequency to detect hacking.

Scope of Statistics in Computer Applications

10. Forecasting and Planning

Statistics helps in predicting future system requirements.

Example :

- Forecasting website traffic for server planning.
- Estimating future data storage requirements.

Scope of Statistics in Computer Applications

Conclusion :

The scope of statistics in computer applications includes data analysis, machine learning, software testing, system performance evaluation, research, security, and decision-making.

Importance of Statistics in Computer Applications



Importance of Statistics in Computer Applications

1. Helps in Data Analysis and Interpretation

Statistics helps computers convert raw data into meaningful information.

Examples:

- Finding average daily users of a website from log data
- Calculating pass percentage of MCA students
- Identifying highest and lowest sales months from data
- Summarizing large datasets into tables and charts

Without statistics, data remains meaningless numbers.

Importance of Statistics in Computer Applications

2. Essential for Machine Learning and Artificial Intelligence

Most AI and ML models are based on statistical concepts.

Examples:

- Classifying emails as spam or non-spam
- Predicting student results using attendance data
- Recommending movies based on user history
- Predicting customer purchase behavior

Statistics explains why a model gives a particular result.

Importance of Statistics in Computer Applications

3. Improves Software Testing and Quality

Statistics helps measure software reliability and performance.

Examples:

- Calculating error rate after testing a program
- Comparing bug frequency in two software versions
- Measuring average execution time of a program
- Analyzing failure rate of a system

Ensures better and more reliable software.

Importance of Statistics in Computer Applications

4. Supports Decision Making in IT Systems

Statistics helps take correct decisions based on data, not guesswork.

Examples:

- Selecting the best algorithm based on performance data
- Choosing a software tool using user ratings
- Deciding server capacity based on traffic data
- Selecting best app version after user testing

Important for managers, developers, and analysts.

Importance of Statistics in Computer Applications

5. Important for Database Management and Data Mining

Statistics helps extract useful patterns from large databases.

Examples:

- Grouping customers into high and low spenders
- Finding frequently accessed database records
- Identifying unusual transactions in banking data
- Analyzing trends from historical records

Used widely in banking, e-commerce, and IT companies.

Importance of Statistics in Computer Applications

6. Helps in Network and System Performance Analysis

Statistics helps monitor and improve system efficiency.

Examples:

- Calculating average server response time
- Measuring internet speed in a computer lab
- Analyzing system downtime frequency
- Identifying peak usage hours

Helps prevent system failure and overload.

Importance of Statistics in Computer Applications

7. Essential for Research and MCA Project Work

Statistics is required to validate project results and conclusions.

Examples:

Comparing two algorithms using test data

Analyzing survey responses for MCA project

Measuring improvement after system optimization

Justifying results in project viva

No project is complete without statistical analysis.

Importance of Statistics in Computer Applications

8. Improves Logical and Analytical Thinking

Statistics develops problem-solving and analytical skills.

Examples:

- Interpreting graphs and reports
- Identifying trends instead of assumptions
- Understanding cause–effect relationships
- Making data-driven arguments in interviews

Very useful for placements and higher studies.

Conclusion

Statistics is important in computer applications for data analysis, machine learning, software quality, decision-making, research, system performance, and logical thinking.

Unit 1- Introduction to Statistics

Types of data:

- Primary and Secondary data,
- Discrete and Continuous data
- Qualitative and Quantitative Data



Primary and Secondary data



What is Primary Data?

When data is collected from beginning to end for the first time by an institution or researcher, such data is called primary data, it is the original data, i.e. the data that is first. It is completely renewed, and it is called primary data. It is collected from scratch a lot of money is spent on collecting primary data. At the same time, manpower is also required, that is, people are also needed.

What is Secondary Data?

Secondary data is data that has already been collected by an institution or researcher and the new researcher generates new information about the use of this already collected data. For example, if you already have the population data of India, then you can easily find out what percentage of people in your city are literate and what percentage of people are illiterate, but you have already collected that data from the population data of India. Therefore, this data will be called Secondary Data.

Primary Data	Secondary Data
Primary data is the first data collected by a researcher for the first time.	Whereas secondary data is a data that is already collected by someone earlier.
Primary data is called real-time data.	While this is not real-time data, it is related to the past.
The process is very much involved in collecting primary data.	While collecting secondary data it does not involve much process but rather quickly and easily.
Primary data is expensive.	While it is economical.
The primary data takes long time for collection.	While secondary data takes shorter time than primary data for collection.
Primary data is available in crude form.	While it is available in processed or refined form.
Primary data is more accurate than secondary data.	While it is less accurate than primary data.
Primary data is more reliable than secondary data.	While secondary data is less reliable than primary data.
There is also difficulty in collecting data.	While there is no difficulty in collecting data.
The sources from which primary data is collected are through surveys, experiments, observations, personal interviews etc.	The sources from which secondary data is collected are websites, government publications, journals, articles etc.
The data is always specific to the researcher's need.	The data collected may or may not be specific to the researcher's need.

Example:

Primary Data	Secondary Data
Log data collected by students while testing a mobile or web application	Application usage statistics taken from company reports or case studies
Survey conducted among MCA students to find preferred programming language (Java, Python, C++)	Programming language usage statistics taken from Stack Overflow Developer Survey

Discrete and Continuous data



Discrete Data Vs. Continuous Data

Discrete Data	Continuous Data
Data that can take only specific, separate values	Data that can take any value within a range
Countable	Measurable
Usually whole numbers	Can be whole numbers, fractions, or decimals
Exact values	Approximate values
Graph Representation -Bar graph	Histogram / Line graph
e.g. Number of errors in a program	e.g. Execution time of a program (in seconds)
e.g. Number of students in a class	e.g. Height or weight of a person

Qualitative and Quantitative data



Qualitative Data Vs. Quantitative Data

Basis	Qualitative Data	Quantitative Data
Meaning	Data that describes qualities or characteristics	Data that represents numerical values
Nature	Descriptive, non-numeric	Numeric, measurable
Type of analysis	Thematic, content analysis	Statistical, mathematical analysis
Answers the question	<i>What type? Which category?</i>	<i>How many? How much?</i>
Collected through	Interviews, observations, open-ended questions	Surveys, experiments, system logs
Mathematical operations	Not suitable for arithmetic operations	Arithmetic operations possible
Graph representation	Pie chart, bar chart (categories)	Histogram, line graph, scatter plot
Example	Type of Software used – System software / Application software / Utility software	Number of errors generated during program execution



Examples

Qualitative Data – (*Descriptive / Categorical* – “What type?”)

- Type of Software used – System software / Application software / Utility software
- User satisfaction level for a mobile app – Very satisfied / Satisfied / Neutral / Unsatisfied
- Error type in a program – Syntax error / Runtime error / Logical error
- Programming paradigm – Object-oriented / Functional / Procedural
- Database type – Relational / NoSQL / Distributed
- Operating system preference – Windows / Linux / macOS
- Website theme – Light mode / Dark mode
- Network type – LAN / MAN / WAN
- Software quality rating – Excellent / Good / Average / Poor
- Type of cyber attack – Phishing / Malware / DDoS

Examples

Quantitative Data (*Numerical / Measurable – “How many? How much?”*)

- Execution time of an algorithm (milliseconds)
- Number of lines of code in a program
- CPU utilization (%)
- Memory usage (MB or GB)
- Number of active users on a website
- Database size (GB)
- Network latency (ms)
- Number of errors generated during program execution
- File download speed (Mbps)
- Response time of a web application (seconds)

Classification of Data

- Types of Data Classification
- Importance of Data Classification



Classification and Presentations of Data

Classification of Data

Classification of data means arranging raw data into groups or categories based on common characteristics so that it becomes easy to understand, analyze, and present.

Presentation of Data

Presentation of data means displaying classified data in a systematic form such as tables, charts, or graphs.

e.g. In computer applications, large data sets are useless unless they are properly classified and presented.

Types of Data Classification

1. Qualitative Classification (Attribute Data)

Meaning:

Data is classified based on qualities or attributes that cannot be measured numerically.

Example :

- Students classified as Hostel / Day Scholar
- Computer systems classified as Desktop / Laptop / Server
- Students classified as Placed / Not Placed

Used when data is descriptive in nature.

Types of Data Classification

2. Quantitative Classification (Numerical Data)

Meaning:

Data is classified based on numerical values that can be measured or counted.

Example:

- Students grouped according to marks in Statistics (0–40, 41–70, 71–100)
- Employees classified by years of experience
- Servers grouped by RAM size (8GB, 16GB, 32GB)
- Internet users grouped by data usage (GB)

Used for mathematical and statistical analysis.

Types of Data Classification

3. Chronological Classification (Time-based Data)

Meaning:

Data is classified according to time such as years, months, days, or hours.

Example:

- Website traffic recorded day-wise
- Software bug reports classified month-wise
- Student admissions recorded year-wise
- System login attempts recorded hour-wise

Helps in studying trends over time.

Types of Data Classification

4. Geographical (Spatial) Classification

Meaning:

Data is classified according to location or place.

Example:

- Number of internet users in different states of India
- IT companies classified by city (Pune, Mumbai, Bengaluru)
- Cybercrime cases recorded state-wise
- Data centers classified by country

Useful for regional comparison and planning.

Importance of Data Classification

1. Improves Data Organization

When data is classified, it becomes easier to store and retrieve information quickly.

Example:

In a **college management system**, student data is classified into:

Personal details

Academic records

Attendance records

Fee information

this classification helps the software retrieve data faster.

Importance of Data Classification

2. Enhances Data Security

Sensitive data needs stronger protection. Classification helps in applying security rules.

Example:

In a **banking application**, data is classified as:

Public Data → Bank branch locations

Confidential Data → Customer account details

Highly Confidential Data → Passwords, OTPs

Strong encryption is applied only to sensitive data.

Importance of Data Classification

3. Helps in Decision Making

Classified data helps companies analyze trends and take business decisions.

Example:

In an **e-commerce application** like Amazon:

Customer purchase data

Product category data

Payment method data

Helps company understand customer behaviour and improve marketing strategies.

Importance of Data Classification

4. Improves Data Analysis and Processing

Classification makes data easier to analyze using software tools.

Example:

In Machine Learning applications:

Data is classified into:

Training Data

Testing Data

Validation Data

Helps improve model accuracy.

Importance of Data Classification

5. Saves Storage and Improves Performance

Not all data needs to be stored at high priority.
Classification helps in storage management.

Example:

In Cloud Computing Systems:

Frequently used data → Stored in fast storage

Old backup data → Stored in archival storage

Improves system performance and reduces cost.

6. Helps in Legal and Compliance Requirements

Companies must protect certain types of data by law.

Example:

In Healthcare Applications:

Patient medical records

Lab reports

Personal identification details

Must follow privacy laws like HIPAA or other regulations.

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- Graphical Presentation of data:
Presentation of Frequency Distribution
by Histogram, Frequency Polygon,
Frequency Curve
 - Diagrammatic Representation using
Multiple and Percentage Bar Diagrams



-Graphical Presentation of data:
Presentation of Frequency Distribution
by Histogram, Frequency Polygon,
Frequency Curve



Histogram

A Histogram is a graphical representation of a continuous frequency distribution in which:

- Class intervals are taken on the X-axis
- Frequencies are taken on the Y-axis
- Rectangles are drawn adjacent to each other (no gaps)
- The area of each rectangle is proportional to the frequency of the class
- Histograms are mainly used when data is grouped into continuous class intervals.

Steps to Draw a Histogram

- Convert data into a continuous frequency distribution
- Mark class intervals on X-axis
- Mark frequencies on Y-axis
- Draw rectangles with width = class interval and height = frequency

Uses of Histogram

1. Performance Analysis in IT Systems

Histograms are used to analyze distribution of system performance metrics such as response time, throughput, or task completion.

Example:

A histogram of API response time shows most requests falling in the 200–300 ms range, helping developers identify acceptable performance limits.

2. Software Quality & Bug Analysis

Used to study frequency of software bugs based on severity or module.

Example:

A histogram of number of bugs per software module helps QA teams identify high-risk modules that require refactoring.

Uses of Histogram

3. Network Traffic Monitoring

Histograms help visualize packet size distribution, latency, or bandwidth usage.

Example:

A network admin plots a histogram of packet delays to detect congestion or abnormal traffic patterns.

4. Database & Query Optimization

Used to analyze query execution time or record distribution in databases.

Example:

A histogram of SQL query execution times highlights slow queries that need indexing or optimization.

Uses of Histogram

5. Cybersecurity & Intrusion Detection

Used to detect anomalies in login attempts or access frequency.

Example:

A histogram showing sudden spikes in failed login attempts indicates a potential brute-force attack.

6. Decision Making in IT Management

Helps managers set benchmarks, thresholds, and SLAs.

Example:

Histogram of tickets resolved per day helps IT managers decide training needs and performance standards.

Frequency Polygon

A Frequency Polygon is a graphical representation of a frequency distribution in which:

- Class midpoints are plotted on the X-axis
- Frequencies are plotted on the Y-axis
- The plotted points are joined by straight lines to form a polygon
- It is mainly used to compare frequency distributions.

Construction of a Frequency Polygon

- Prepare a grouped frequency distribution
- Calculate midpoints of each class interval
- Plot midpoints on X-axis and frequencies on Y-axis
- Join the points with straight lines
- Extend the polygon to meet the X-axis at both ends

Uses of Frequency Polygon

1. Comparison of Performance Distributions

Frequency polygons are widely used to compare two or more continuous datasets on the same graph.

Example:

Comparing frequency polygons of ticket resolution rates of two IT support teams helps identify which team performs better overall.

2. Trend Analysis in System Performance

Used to study increasing or decreasing trends in technical data.

Example:

A frequency polygon of daily server response times shows performance improvement or degradation over time.

Uses of Frequency Polygon

3. Network Traffic Analysis

Helps visualize and compare network latency or packet transmission patterns.

Example:

Frequency polygons of packet delay before and after network optimization show the effectiveness of changes.

4. Software Quality Comparison

Used to compare bug frequency across software versions or modules.

Example:

A frequency polygon comparing number of defects in Version 1 vs Version 2 highlights quality improvements.

Uses of Frequency Polygon

5. Database Query Performance Evaluation

Used to compare query execution time distributions.

Example:

Frequency polygons of SQL query execution times before and after indexing show performance gains.

7. Decision Support in IT Management

Supports data-driven decisions related to benchmarking and optimization.

Example:

Frequency polygon of employees' productivity levels helps management set realistic performance standards.

Conclusion

Frequency polygons are powerful graphical tools in computer applications for comparing distributions, analyzing trends, and supporting technical and managerial decisions in IT environments.

Frequency Curve

- A Frequency Curve is a smooth free-hand curve drawn by joining the points of a frequency polygon. It represents the overall pattern of a frequency distribution and shows how data is distributed over different class intervals.
- A frequency curve is an effective graphical tool for representing continuous data, helping in trend analysis, comparison, and decision making in statistical and technical applications.

Construction of a Frequency Curve

- Prepare a grouped frequency distribution
- Calculate class midpoints
- Plot midpoints on the X-axis and frequencies on the Y-axis
- Join the plotted points smoothly (not with straight lines)
- Ensure the curve starts and ends on the X-axis

Uses of frequency Curve

1. Trend Analysis in Performance Data

Helps in understanding overall trend and behavior of large datasets.

Example:

A frequency curve of daily CPU utilization shows whether system load is generally stable or fluctuating.

2. Prediction and Forecasting

Smooth curves help in estimating future values and performance expectations.

Example:

A frequency curve of monthly software tickets helps predict future support workload.

Uses of frequency Curve

3. Identifying Concentration of Data

Shows where data values are most densely concentrated.

Example:

Frequency curve of query execution time shows most queries executed within a particular time range.

4. Quality Control & Process Monitoring

Used to monitor process stability and detect deviations.

Example:

A frequency curve of defect resolution time helps QA teams detect abnormal process variations.

Uses of frequency Curve

3. Decision Support in IT Management

Supports managerial decisions by visualizing overall efficiency trends.

Example:

Frequency curve of employee productivity helps management decide on training and resource allocation.

Conclusion

A frequency curve is an effective graphical tool for representing continuous data, helping in trend analysis, forecasting, quality control, and performance evaluation in computer applications.

Example 1

An IT company **Infosys** conducted an internal **coding assessment** for MCA trainees.

Marks obtained by **50 trainees** are grouped as follows:

Coding Score (Class Interval)	Number of Trainees
0 – 10	5
10 – 20	9
20 – 30	14
30 – 40	12
40 – 50	10



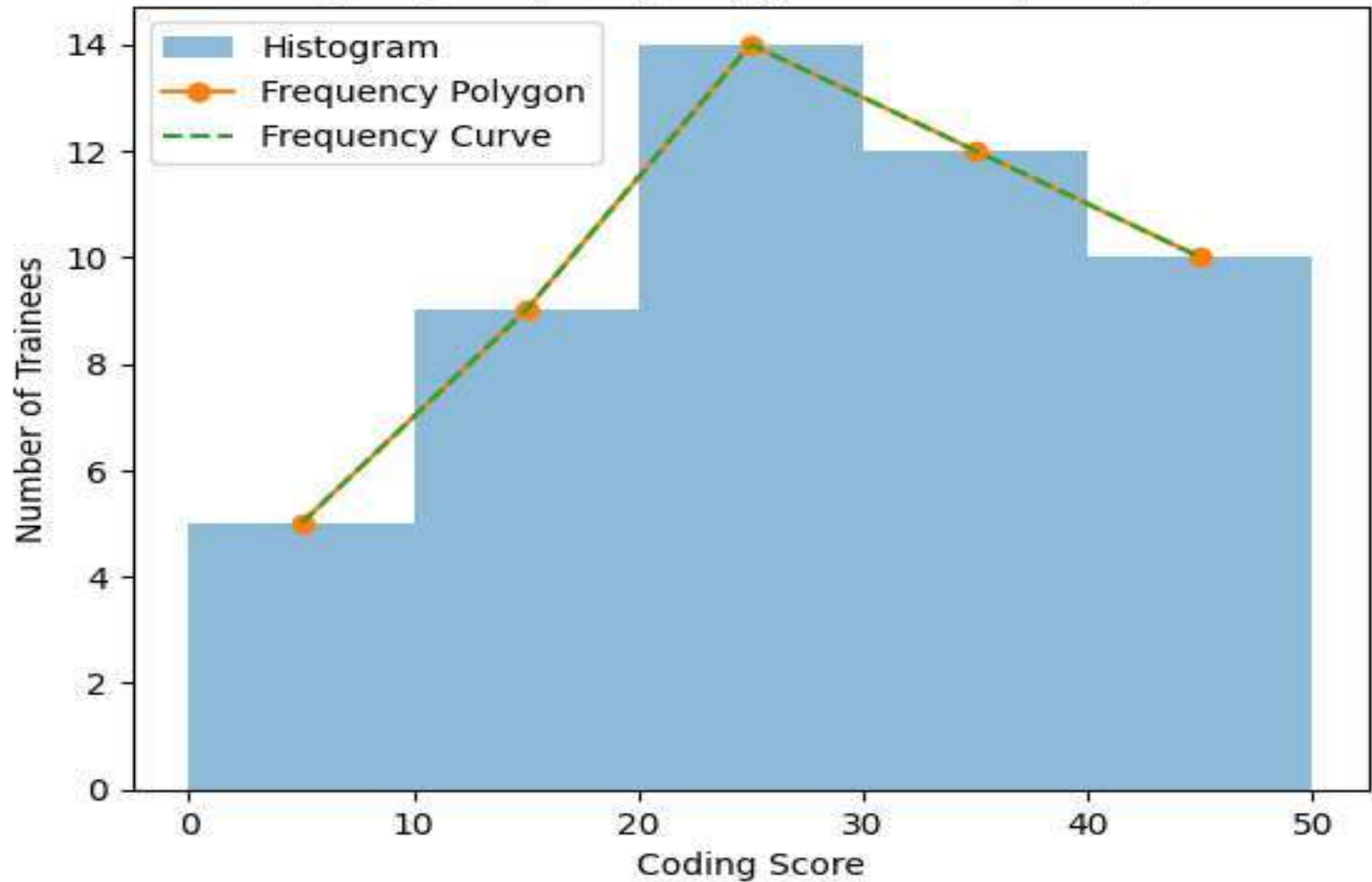
Questions

Q. 1 Represent the above data using a Histogram, Frequency Polygon and Frequency Curve.

Q.2 The highest number of trainees scored between 20–30 marks. As a training manager at Infosys, what decision should be taken to improve overall coding performance?

Q. 3 Analysis shows that 14 trainees scored below 20 marks. What is the best managerial decision Infosys should take?

Histogram, Frequency Polygon and Frequency Curve



Q2. The highest number of trainees scored between **20–30 marks**.

As a training manager at Infosys, what **decision** should be taken to improve overall coding performance?

Expected Answer (Solution):

- Identify weak coding concepts from assessment
- Introduce focused re-training modules
- Use AI-based personalized learning paths
- Conduct regular low-stakes coding tests

Decision based on performance distribution

Q. 3 Analysis shows that 14 trainees scored below 20 marks. What is the best managerial decision Infosys should take?

Answer:

Provide mentoring and extra practice sessions

Justification:

Reduces trainee failure rate

Improves workforce quality

Saves recruitment and onboarding cost

Data-driven HR decision

Example 2

Wipro wants to analyse **daily ticket resolution performance** of its support engineers before de promotions and training.

Number of tickets resolved per day by **100 employees**:

Tickets Resolved per Day	Number of Employees
0 – 5	8
5 – 10	12
10 – 15	20
15 – 20	30
20 – 25	18
25 – 30	12



Questions

Q. 1 Represent the above data using a Histogram, Frequency Polygon and Frequency Curve.

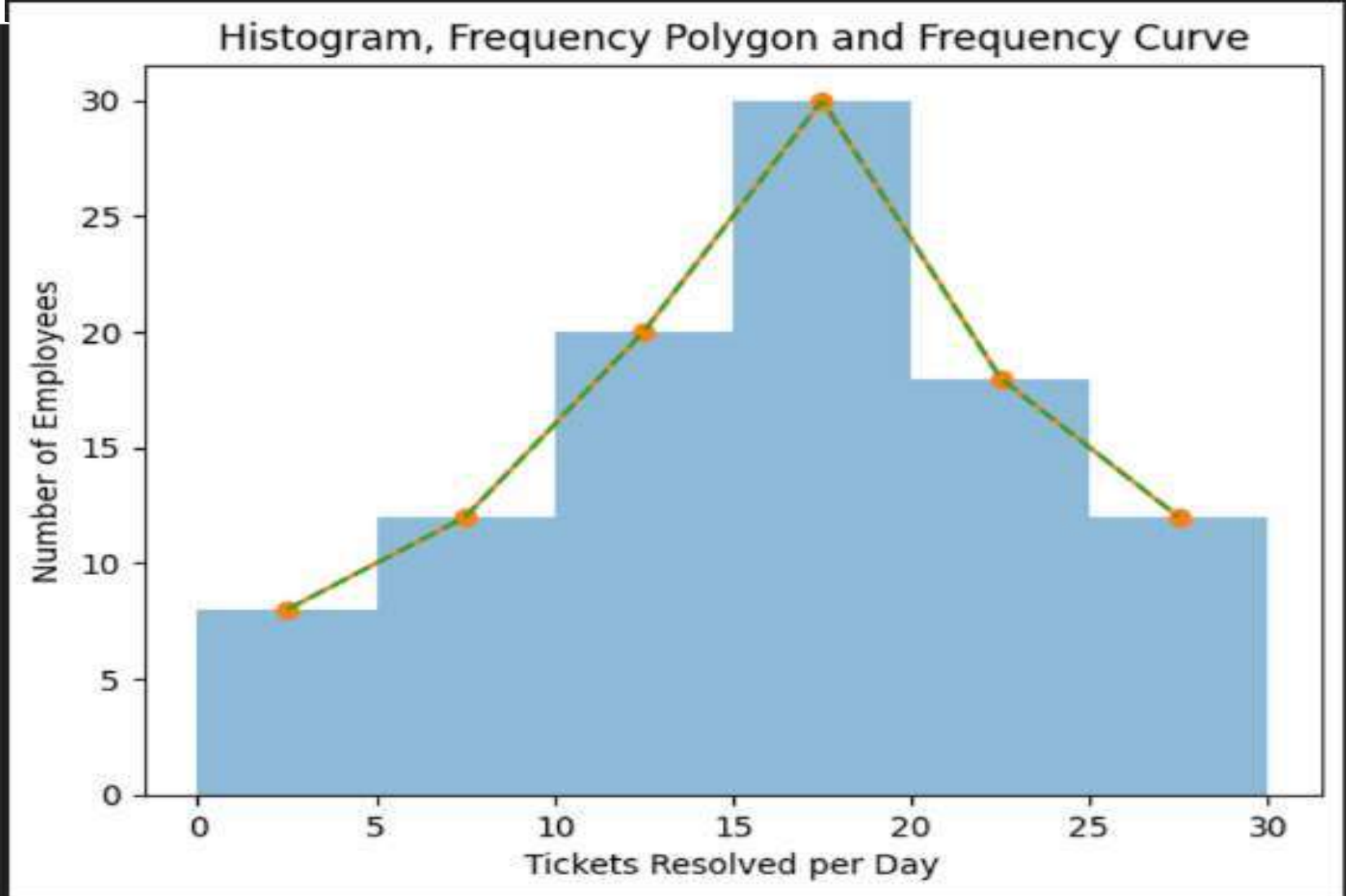
Q. 2 Wipro wants to identify employees who need immediate training due to low productivity. Which ticket-resolution group should be prioritized for training programs, and why?

Q. 3 If Wipro plans to promote only top-performing engineers, which ticket-resolution category should be considered for promotion and why?

Q. 4 Management wants to maintain productivity with minimal training cost. Which group of employees should be used as mentors or peer trainers for others?

Q. 5 Wipro plans to set a minimum daily performance benchmark for engineers. Based on the data, what should be a reasonable benchmark, and why?

Q. 1



Answer :

Q. 2 Wipro wants to identify employees who need immediate training due to low productivity. Which ticket-resolution group should be prioritized for training programs, and why?

Answer:

Employees resolving 0–5 and 5–10 tickets/day.

Justification:

0–5 → 8 employees

5–10 → 12 employees

Total low performers = 20 employees (20%)

These employees are significantly below the average productivity range and would benefit most from skill enhancement and workflow training.

Answer :

Q. 3 If Wipro plans to promote only top-performing engineers, which ticket-resolution category should be considered for promotion and why?

Answer:

Employees resolving 20–25 and 25–30 tickets/day.

Justification:

20–25 → 18 employees

25–30 → 12 employees

These employees show high efficiency, experience, and problem-solving speed, making them ideal candidates for senior roles or team leads.

Answer :

Q. 4 Management wants to maintain productivity with minimal training cost. Which group of employees should be used as mentors or peer trainers for others?

Answer:

Employees resolving 15–20 tickets/day.

Justification:

Largest group: 30 employees (30%)

They represent average-to-good performance

Can bridge the gap between low and high performers by mentoring without additional hiring cost.

Answer :

Q. 5 Wipro plans to set a minimum daily performance benchmark for engineers. Based on the data, what should be a reasonable benchmark, and why?

Answer:

10 tickets per day.

Justification:

Employees resolving ≥ 10 tickets/day = 80% of workforce

Setting the benchmark at 10 tickets:

- Is realistic

- Avoids demotivation

- Encourages low performers to improve without penalizing the majority

- Diagrammatic Representation using Multiple and Percentage Bar Diagrams



Syllabus

- Diagrammatic Representation using Multiple Bar Diagrams



Multiple Bar Diagram

A Multiple Bar Diagram (also called a Grouped Bar Diagram) is a graphical representation in which two or more related variables are shown by sets of bars placed side by side for the same category.

It is mainly used for comparison of related datasets.

Construction of a Multiple Bar Diagram

- Collect data for two or more variables related to the same categories
- Draw categories on the X-axis
- Draw values on the Y-axis
- For each category, draw bars side by side, each bar representing a different variable
- Use different colors or patterns and provide a legend

Uses of Multiple Bar Diagram

1. Comparison of Performance Metrics

Used to compare multiple performance parameters for the same system or department.

Example:

Comparing CPU usage, memory usage, and disk usage of two servers.

2. Software Version Comparison

Used to compare performance of different software versions.

Example:

Multiple bar diagram showing execution time, memory consumption, and error rate of Version 1 and Version 2 of a software.

Uses of Multiple Bar Diagram

3. Team or Department Comparison

Used to compare productivity of different teams.

Example:

Comparing tickets resolved by Team A and Team B over different days.

4. Database Performance Evaluation

Used to compare database efficiency metrics.

Example:

Comparing query execution time across different database engines (MySQL vs PostgreSQL).

Uses of Multiple Bar Diagram

5. Decision Making in IT Management

Helps management in resource allocation and benchmarking.

Example:

Comparing training cost vs productivity output of different employee groups.

Conclusion

A multiple bar diagram is an effective graphical tool for comparing related datasets and is widely used in computer applications for performance analysis, system evaluation, and managerial decision making.

Example 1

A mid-size IT company analyzed its **IT infrastructure usage** over three years to decide between **Cloud Computing** and **On-Premise Systems**.

The percentage usage of each infrastructure is shown below:

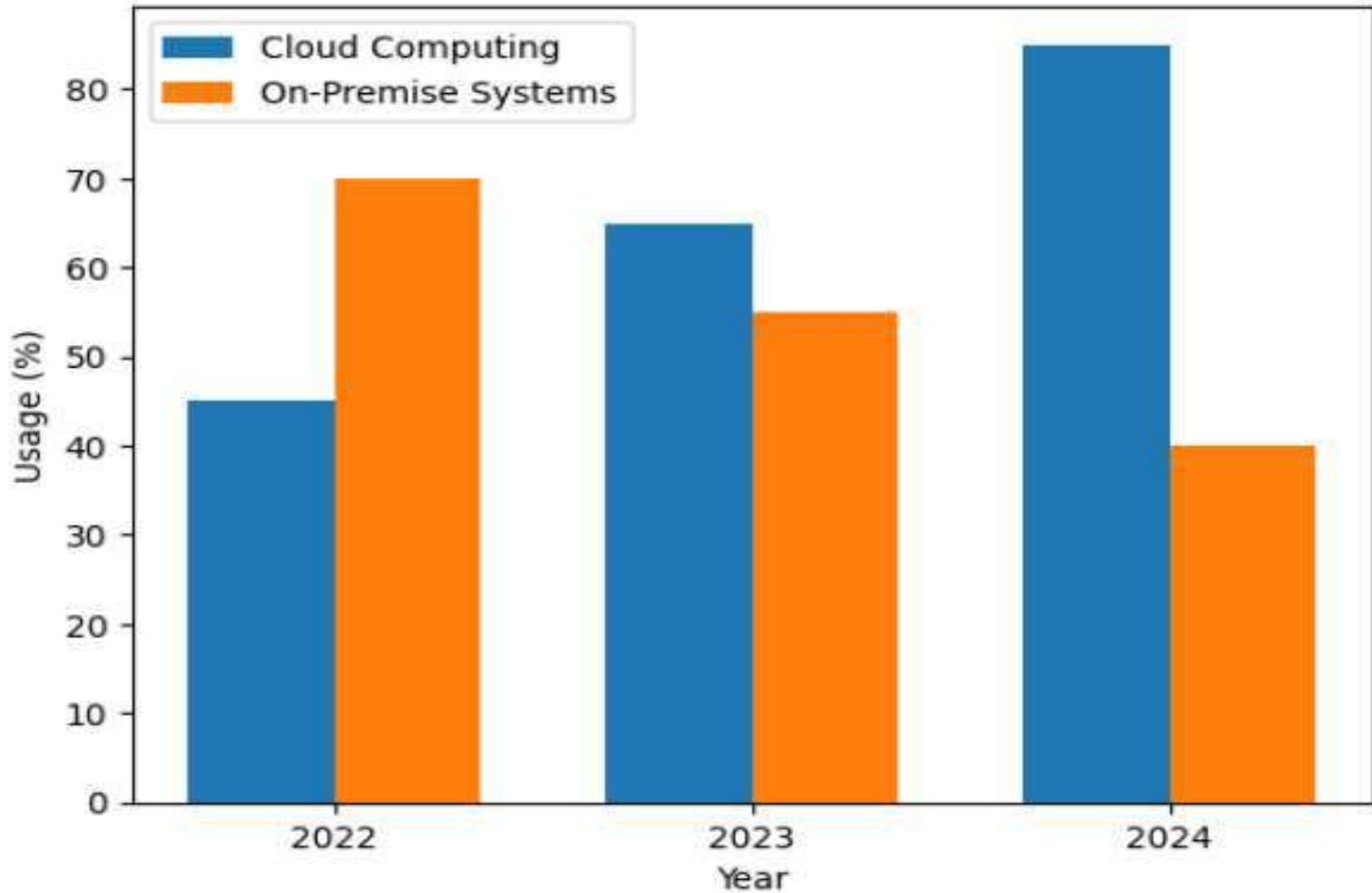
Year	Cloud Computing (%)	On-Premise Systems (%)
2022	45	70
2023	65	55
2024	85	40

Question :

1. Represent the above data using a Multiple Bar Diagram.
2. Which computing model is gaining importance in the company?
3. In which year does Cloud Computing overtake On-Premise systems?
4. As a CIO, which infrastructure should be prioritized for future investment and why?
5. Based on the trend shown in the diagram, should the CIO move all systems completely to Cloud or adopt a Hybrid model? Justify your answer.



Multiple Bar Diagram: IT Infrastructure Usage in a Company



Answer :

Q. 2. Which computing model is gaining importance in the company?

Decision : Cloud Computing is gaining importance, as its usage increases consistently every year.

Q. 3. In which year does Cloud Computing overtake On-Premise systems?

Decision : Cloud Computing overtakes On-Premise systems in **2023**.

Q. 4. As a CIO, which infrastructure should be prioritized for future investment and why?

Decision : The company should prioritize Cloud Computing because it shows rapid adoption, scalability, and reduced dependence on traditional infrastructure.

Answer :

Q. 5. Based on the trend shown in the diagram, should the CIO move all systems completely to Cloud or adopt a Hybrid model? Justify your answer.

Decision :

The CIO should adopt a Hybrid Infrastructure model.

Justification:

Cloud Computing shows rapid growth and scalability, making it suitable for high availability and flexible workloads.

On-Premise systems, although declining, still offer better control and compliance for sensitive data.

A hybrid model balances performance, security, and regulatory requirements while minimizing risk.

Example 2

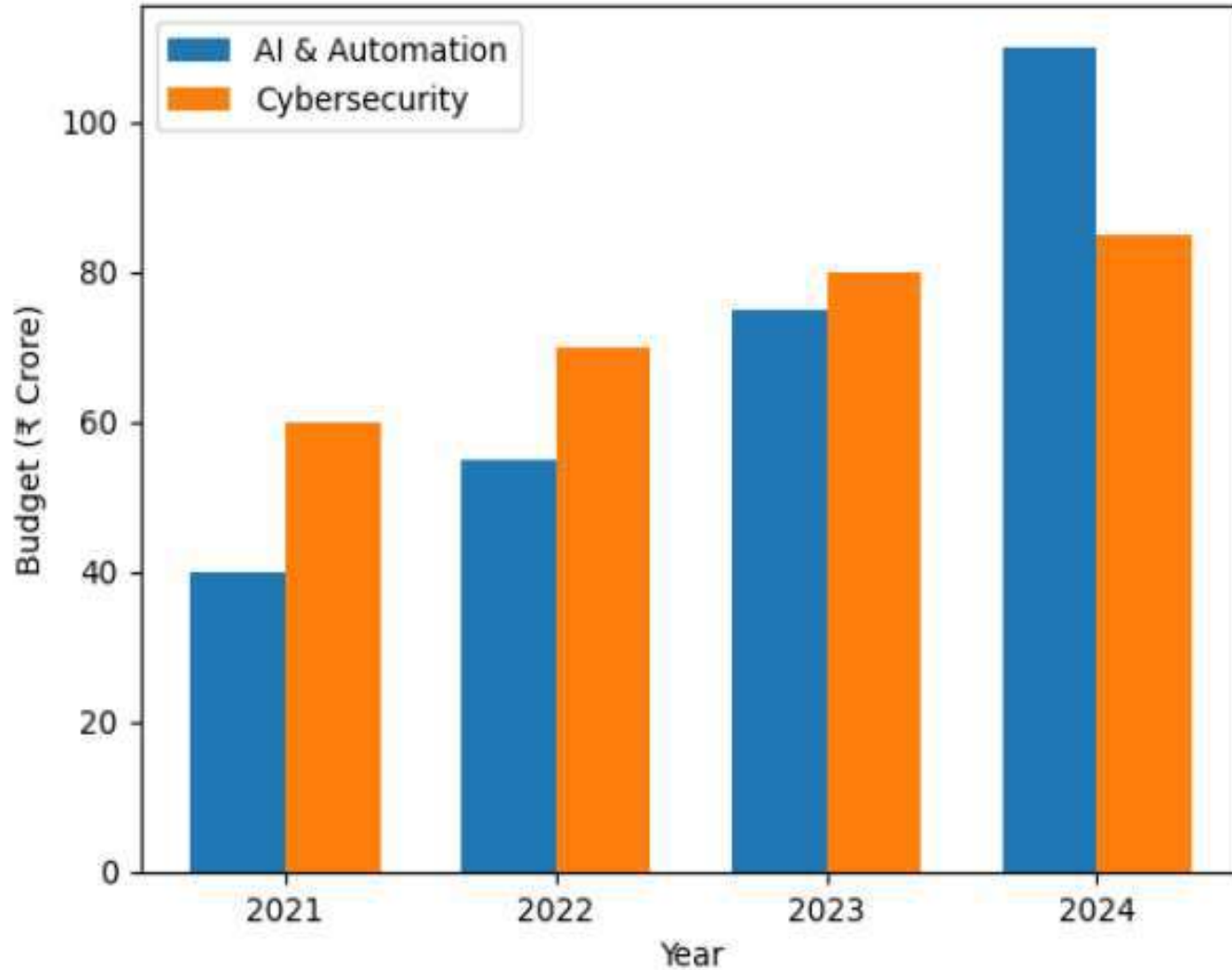
A multinational IT company analyzed its **annual IT budget allocation** (₹ in crore) between **AI & Automation** and **Cybersecurity** to support digital transformation and data protection.

Year	AI & Automation (₹ Cr)	Cybersecurity (₹ Cr)
2021	40	60
2022	55	70
2023	75	80
2024	110	85

Question

1. Represent the above data using a Multiple Bar Diagram.
2. In which year does the company's strategic focus shift from security-driven IT spending to innovation-driven IT spending? Justify your answer using data.
3. Despite continuous growth in Cybersecurity spending, why might the management still consider AI & Automation as a higher-priority investment after 2023?
4. Does the diagram indicate a risk-reward trade-off in the company's IT strategy?

Annual IT Budget Allocation



Answer:

Q. 2 In which year does the company's strategic focus shift from security-driven IT spending to innovation-driven IT spending? Justify your answer using data.

Answer:

From 2021–2023, Cybersecurity spending is higher than AI & Automation:

2021: AI = 40, Cyber = 60

2022: AI = 55, Cyber = 70

2023: AI = 75, Cyber = 80

In 2024, AI & Automation jumps to ₹110 Cr, overtaking Cybersecurity at ₹85 Cr.

Interpretation:

This crossover indicates a strategic shift from a security-driven approach (risk protection) to an innovation-driven approach (growth, efficiency, digital transformation).

Q. 3 Despite continuous growth in Cybersecurity spending, why might the management still consider AI & Automation as a higher-priority investment after 2023?

Decision :

Even though Cybersecurity spending continues to increase, AI & Automation offers higher strategic returns.

Key reasons:

AI & Automation directly drives revenue growth through:

- Process automation

- Predictive analytics

- Cost reduction

- Faster decision-making

Cybersecurity is largely a protective investment (prevents loss, but doesn't generate new value).

After 2023, the company likely achieved a baseline security maturity, allowing it to:

- Shift incremental investment toward innovation
- Use AI to improve even security operations (AI-based threat detection)

Conclusion :

Management prioritizes AI & Automation because it delivers scalable competitive advantage, whereas Cybersecurity mainly ensures stability.

Answer

Q. 4 Does the diagram indicate a risk-reward trade-off in the company's IT strategy?

Answer :

Yes, the diagram clearly indicates a risk-reward trade-off.

Explanation:

AI & Automation = High reward, higher risk

Drives innovation, speed, efficiency

Cybersecurity = Lower reward, risk mitigation

Ensures data integrity, uptime, trust

- Diagrammatic Representation using Percentage Bar Diagrams



Percentage Bar Diagram

- A Percentage Bar Diagram is a graphical representation in which:
- Each bar represents a total value of 100%
- The bar is divided into components expressed as percentages
- It shows the relative contribution of each component to the total
- This diagram focuses on proportions rather than absolute values.

Construction of a Percentage Bar Diagram

- Convert all data values into percentages
- Draw bars of equal height or length (100%)
- Divide each bar according to the percentage values
- Use different shading or colors for each component

Uses of Percentage Bar Diagram

1. Resource Utilization Analysis

Used to show percentage usage of system resources.

Example:

A percentage bar diagram representing CPU (45%), Memory (35%), Disk (20%) usage of a server.

2. Software Development Effort Distribution

Used to analyze percentage contribution of each SDLC phase.

Example:

Design – 30%, Coding – 40%, Testing – 20%, Maintenance – 10%.

Uses of Percentage Bar Diagram

3. Database Storage Usage

Used to compare percentage allocation of database components.

Example:

Tables – 60%, Indexes – 25%, Logs – 15%.

4. Software Quality & Defect Analysis

Used to analyze types of bugs contributing to failures.

Example:

UI bugs – 20%, Logic bugs – 40%, Performance bugs – 25%, Security bugs – 15%.

5. Performance Contribution Analysis

Used to show relative impact of different factors on performance.

Example:

CPU delay – 50%, Network delay – 30%, I/O delay – 20%.

Uses of Percentage Bar Diagram

6. IT Management & Decision Making

Helps managers understand where maximum resources or time are spent.

Example:

Percentage bar diagram showing department-wise IT budget utilization.

Conclusion

Percentage bar diagrams are widely used in computer applications to represent proportional data clearly, enabling effective analysis of resource usage, performance contribution, and decision making in IT environments.

Example 1

A software company completed projects using different programming languages:

Language	Number of Projects
Python	70
Java	50
JavaScript	40
C++	20
Others	20

Questions:

Q. 1 Draw a percentage bar diagram showing distribution of projects by programming language.

Q2. Which programming language should the company prioritize for hiring new developers?

Q3. If company wants to reduce dependency on a single language, which language usage should they reduce?

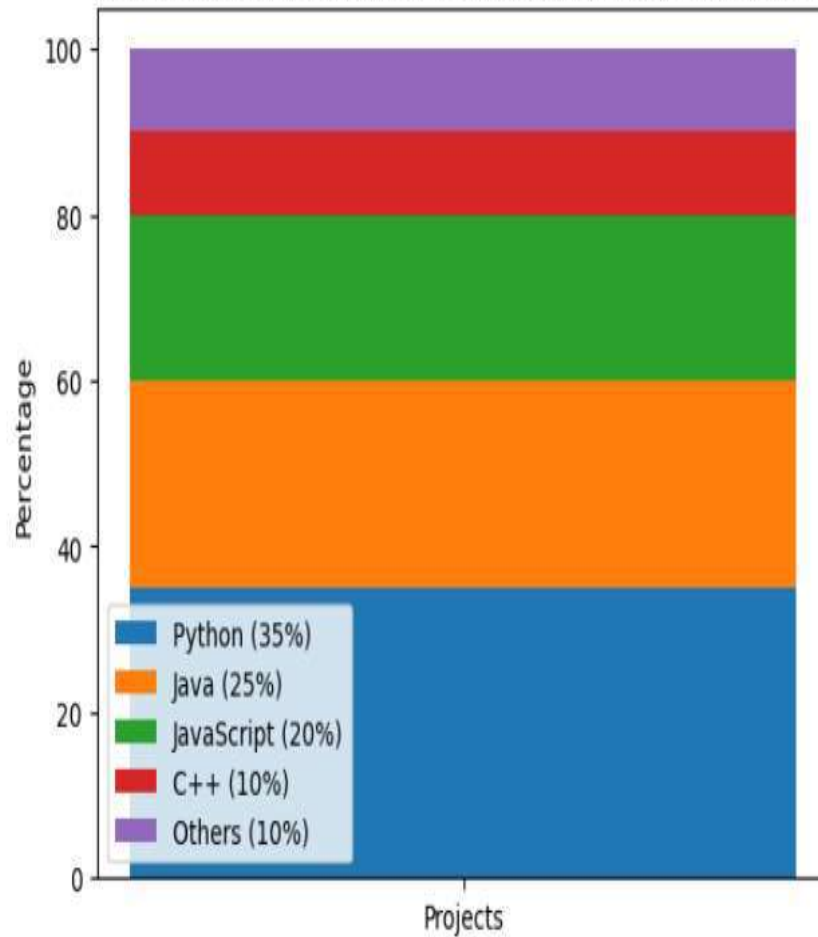
Q4. If demand for web development increases, which language category should expand?

Percentage Conversion Table

Language	Number of Projects	Percentage
Python	70	35%
Java	50	25%
JavaScript	40	20%
C++	20	10%
Others	20	10%
Total	200	100%



Percentage Bar Diagram of Programming Language Projects



Answer

Q2. Which programming language should the company prioritize for hiring new developers?

Answer: Python (highest project share – 35%)

Q3. If company wants to reduce dependency on a single language, which language usage should they reduce?

Answer: Python, since it has the highest dependency.

Q4. If demand for web development increases, which language category should expand?

Answer: JavaScript (mainly used for web technologies).

Example 2

Example : Software Modules Developed in a Company

Module Type	Number of Modules
Web Applications	80
Mobile Applications	60
Desktop Applications	40
AI/ML Projects	20

Questions

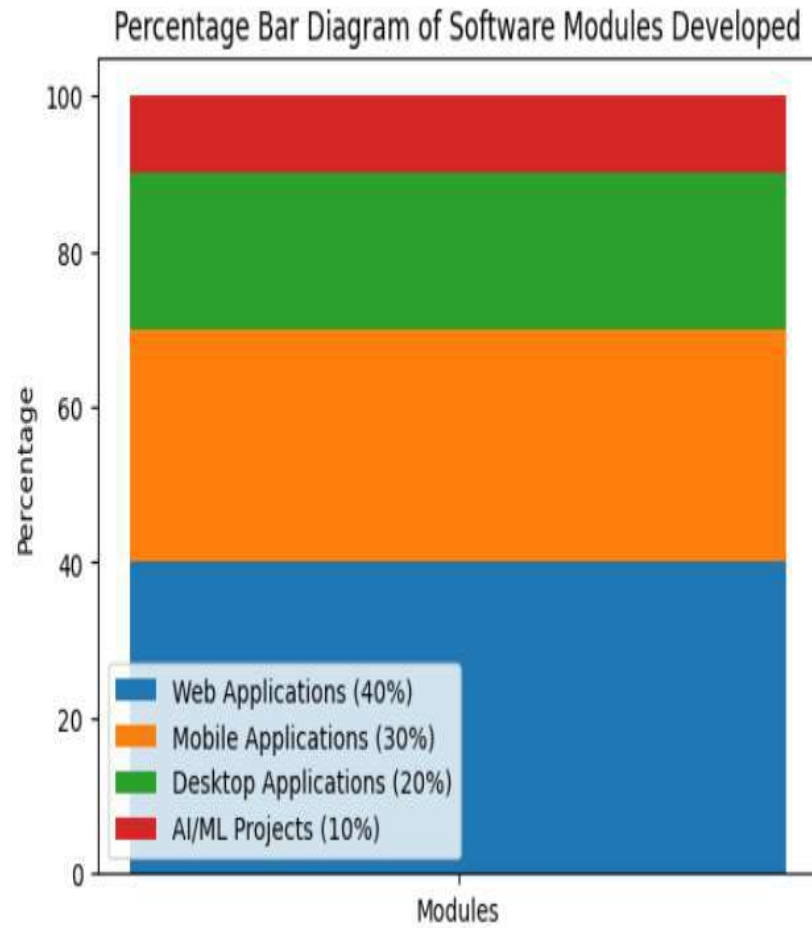
- Q. 1 Draw percentage bar diagram.
- Q. 2 Which module type represents the company's core business focus, and where should maximum hiring be done?
- Q. 3 If the company wants to expand into future-oriented technologies, which module type should receive more investment and training?
- Q. 4 If market demand for smartphone-based services increases rapidly, which module type should the company prioritize for development?
- Q. 5 If company resources are limited and management wants to reduce work in declining technology areas, which module type is most suitable for reduction?
- Q.6 If the company wants to maintain balanced expertise across platforms (web, mobile, desktop, AI/ML), which module type needs the highest growth effort?

Answer:

Percentage Conversion Table

Module Type	Number of Modules	Percentage
Web Applications	80	40%
Mobile Applications	60	30%
Desktop Applications	40	20%
AI/ML Projects	20	10%
Total	200	100%





Answer :

Q. 2 Which module type represents the company's core business focus, and where should maximum hiring be done?

Answer : Web Applications (40%)

Because it has the highest share, showing it is the main working area.

Maximum hiring should be done in Web Application development.

Q. 3 If the company wants to expand into future-oriented technologies, which module type should receive more investment and training?

Answer : AI/ML Projects (10%)

AI and ML are future technologies and currently have the lowest share, so they need expansion.



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Answer

Q4. If market demand for smartphone-based services increases rapidly, which module type should the company prioritize for development?

Answer : Mobile Applications (30%)

Smartphone services mainly depend on mobile app development.

Q5. If company resources are limited and management wants to reduce work in declining technology areas, which module type is most suitable for reduction?

Answer : Desktop Applications (20%)

Desktop software demand is generally declining compared to web and mobile platforms.

Answer

Q6. If the company wants to maintain balanced expertise across platforms (web, mobile, desktop, AI/ML), which module type needs the highest growth effort?

Answer : AI/ML Projects (10%)

Because it has the lowest percentage and needs growth to match other fields.



Thank You



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